



QUESTION BANK

Details of the Course

Name of the Department	: Computer Science and Engineering
Name of the Course	: Theory of Computation
Course Code	: 23CS1502
Regulation	: R2023
Semester	: V
Common To Programme(s)	: Information Technology

Instructions

- Questions must be clear, unambiguous, and error-free.**

- Avoid repetitive or overlapping questions.
- Use appropriate diagrams/tables wherever required.
- Numerical problems must include **complete data**.
- Avoid outdated or syllabus-irrelevant topics.

Blooms Level: Blooms Level 1 & 2 is Lower Order (LO) Cognitive type, Blooms Level 3 & 4 is Intermediate Order Cognitive Type (IO) and Blooms Level 5 & 6 is Higher Order (HO) cognitive type.

Course Outcome: (List the Course Outcomes of the Course)

- CO1:** Construct finite automata, regular expression for any pattern.
CO2: Write context free grammar for any construct.
CO3: Build pushdown automata to recognize a context free language.
CO4: Design Turing machines for any language.
CO5: Propose computation solutions using Turing Machine.
CO6: Drive whether a problem is decidable or not.

Bloom's Level: **BL1** - Remembering, **BL2** - Understanding, **BL3** - Applying, **BL4** - Analyzing, **BL5** - Evaluating, **BL6** - Creating.

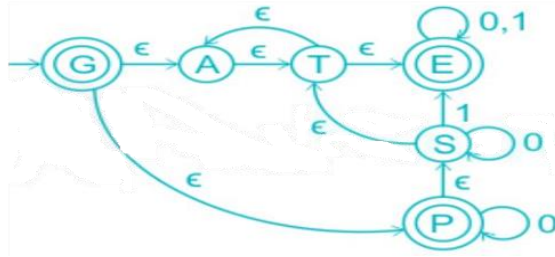
UNIT- I FINITE AUTOMATA

Questions should be of:

- Basic concepts, definitions, laws, and simple applications.
- Lower order (LO) cognitive and may be of Intermediate order (IO) cognitive type.

PART A (2 Marks)		Bloom's Level	Course Outcome	Marks Allotted
1.	State the principle of mathematical induction.	[BL1]	[CO1]	[2]
2.	Prove $1+2+3+\dots+k=k(k+1)/2$ using induction method.	[BL2]	[CO1]	[2]
3.	If p is prime the $M_p=2^p-1$ is prime. Give its counter example.	[BL2]	[CO1]	[2]
4.	Differentiate between proof by contradiction and proof by contra positive.	[BL2]	[CO1]	[2]
5.	Define finite automata.	[BL1]	[CO1]	[2]
6.	List the components of DFA.	[BL1]	[CO1]	[2]
7.	State the applications and limitations of Finite Automata	[BL1]	[CO1]	[2]
8.	Design FA with $\Sigma = \{0, 1\}$ accepts the set of all strings with three consecutive 0's.	[BL2]	[CO1]	[2]
9.	Compare DFA, NFA, and ϵ -NFA.	[BL2]	[CO1]	[2]

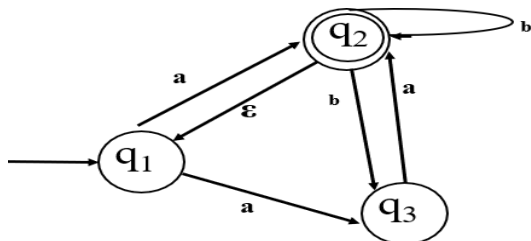
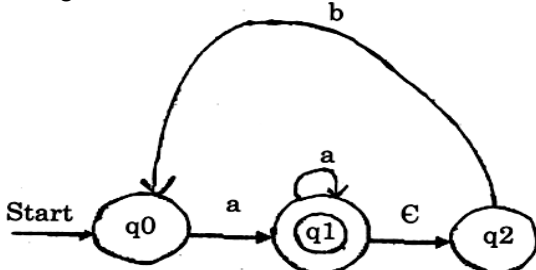
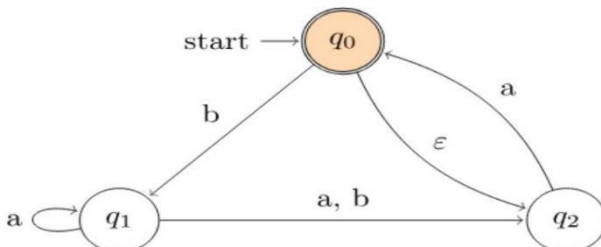
10.	Find ϵ -closure for the following non-deterministic finite automata (NFA) with epsilon.	[BL2]	[CO1]	[2]
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PART B (13 Marks)

- i. Questions should be of Concept explanation, detailed reasoning, numerical problems, and derivations.
 ii. **Questions should be of Understanding / Applying/ Analyzing/ Evaluating**
 iii. Maximum number of Subdivisions is **Three**.

1.	i. Prove the following by the principle of induction. $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$ ii. Prove that $\sqrt{2}$ is not rational.	[BL2]	[CO1]	[7]															
2.	i. Given $\Sigma = \{a, b\}$ construct a DFA which recognize the language $L = \{b^m a b^n : m, n > 0\}$. ii. Construct the DFA accepting the language over the alphabet $\Sigma = \{0, 1\}$, where number of 1's is a multiple of 3.	[BL5]	[CO1]	[7]															
3.	Construct NFA that accepts all string that ends in 01. Give its transition table and extended transition function for the input string 00101 also construct a DFA for the above NFA using subset construction method.	[BL3]	[CO1]	[13]															
4.	Analyze the procedure for converting ϵ -NFA into DFA with a suitable example.	[BL4]	[CO1]	[13]															
5.	Construct an NFA accepting the set of strings over $\{a, b\}$ ending in aba, use it to construct a DFA accepting the same set of strings	[BL3]	[CO1]	[13]															
6.	Construct DFA equivalent to NFA $(\{P, Q, R, S\}, \{0, 1\}, \delta, P, \{S\})$, Where δ is defined as <table><tr><td>δ</td><td>0</td><td>1</td></tr><tr><td>P</td><td>$\{P, Q\}$</td><td>$\{P\}$</td></tr><tr><td>Q</td><td>$\{R\}$</td><td>$\{R\}$</td></tr><tr><td>R</td><td>$\{S\}$</td><td>-</td></tr><tr><td>S</td><td>$\{S\}$</td><td>$\{S\}$</td></tr></table>	δ	0	1	P	$\{P, Q\}$	$\{P\}$	Q	$\{R\}$	$\{R\}$	R	$\{S\}$	-	S	$\{S\}$	$\{S\}$	[BL5]	[CO1]	[13]
δ	0	1																	
P	$\{P, Q\}$	$\{P\}$																	
Q	$\{R\}$	$\{R\}$																	
R	$\{S\}$	-																	
S	$\{S\}$	$\{S\}$																	
7.	Convert the following NFA to DFA. <table><tr><td>δ</td><td>A</td><td>B</td></tr><tr><td>$\rightarrow P$</td><td>$\{p\}$</td><td>$\{p, q\}$</td></tr><tr><td>Q</td><td>$\{r\}$</td><td>$\{r\}$</td></tr><tr><td>$*r$</td><td>$\{\phi\}$</td><td>$\{\phi\}$</td></tr></table>	δ	A	B	$\rightarrow P$	$\{p\}$	$\{p, q\}$	Q	$\{r\}$	$\{r\}$	$*r$	$\{\phi\}$	$\{\phi\}$	[BL5]	[CO1]	[13]			
δ	A	B																	
$\rightarrow P$	$\{p\}$	$\{p, q\}$																	
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$*r$	$\{\phi\}$	$\{\phi\}$																	

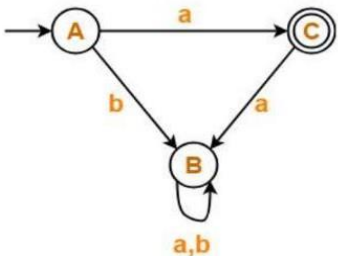
8.	Convert the ϵ -NFA to DFA and list the difference between NFA and DFA.	[BL4]	[CO1]	[13]																		
																						
9.	Convert the following ϵ –NFA to NFA and then convert the resultant NFA to DFA.	[BL5]	[CO1]	[13]																		
																						
10.	Construct an NFA without ϵ -transition for the NFA give below.	[BL5]	[CO1]	[13]																		
																						
Part C (15 Marks) (Only of Higher Order Thinking Question)																						
i. Questions should be of Higher Order (HO) cognitive type.																						
ii. Formulate questions that align with the Evaluate and Create levels of higher-order thinking.																						
iii. Maximum number of Subdivisions is Three .																						
1.	Prove that if n is a positive integer such that $n \bmod 4$ is 2 or 3 then n is not a perfect square.	[BL5]	[CO1]	[15]																		
2.	Give the DFA accepting the following languages over the alphabet i. Number of 0's is a multiple of 3 ii. Number of 0's is not a multiple of 3	[BL5]	[CO1]	[15]																		
3.	Convert the following NFA to a DFA and informally describe the language it accepts.	[BL5]	[CO1]	[15]																		
<table><tr><td>δ</td><td>0</td><td>1</td></tr><tr><td>$\rightarrow p$</td><td>{p,q}</td><td>{p}</td></tr><tr><td>q</td><td>{r,s}</td><td>{t}</td></tr><tr><td>r</td><td>{p,r}</td><td>{t}</td></tr><tr><td>*s</td><td>ϕ</td><td>ϕ</td></tr><tr><td>*t</td><td>ϕ</td><td>ϕ</td></tr></table>					δ	0	1	$\rightarrow p$	{p,q}	{p}	q	{r,s}	{t}	r	{p,r}	{t}	*s	ϕ	ϕ	*t	ϕ	ϕ
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*s	ϕ	ϕ																				
*t	ϕ	ϕ																				

4.	Consider the following ϵ-NFA. Compute ϵ – closure of each state and find its equivalent DFA. <table border="1"> <thead> <tr> <th>δ</th><th>ϵ</th><th>a</th><th>b</th><th>c</th></tr> </thead> <tbody> <tr> <td>$\rightarrow p$</td><td>$\{q, r\}$</td><td>$\{\varnothing\}$</td><td>$\{q\}$</td><td>$\{r\}$</td></tr> <tr> <td>q</td><td>$\{\varnothing\}$</td><td>$\{p\}$</td><td>$\{r\}$</td><td>$\{p, q\}$</td></tr> <tr> <td>$*r$</td><td>$\{\varnothing\}$</td><td>$\{\varnothing\}$</td><td>$\{\varnothing\}$</td><td>$\{r\}$</td></tr> </tbody> </table>	δ	ϵ	a	b	c	$\rightarrow p$	$\{q, r\}$	$\{\varnothing\}$	$\{q\}$	$\{r\}$	q	$\{\varnothing\}$	$\{p\}$	$\{r\}$	$\{p, q\}$	$*r$	$\{\varnothing\}$	$\{\varnothing\}$	$\{\varnothing\}$	$\{r\}$	[BL6]	[CO1]	[15]
δ	ϵ	a	b	c																				
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q	$\{\varnothing\}$	$\{p\}$	$\{r\}$	$\{p, q\}$																				
$*r$	$\{\varnothing\}$	$\{\varnothing\}$	$\{\varnothing\}$	$\{r\}$																				
5.	Convert the following ϵ –NFA to NFA and then convert the resultant NFA to DFA.	[BL6]	[CO1]	[15]																				

UNIT- II REGULAR EXPRESSION AND REGULAR LANGUAGES

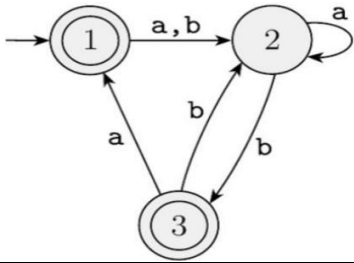
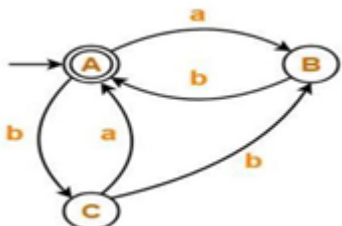
Questions should be of:

- i. Basic concepts, definitions, laws, and simple applications.
- ii. Lower order (LO) cognitive and may be of Intermediate order (IO) cognitive type.

PART A (2 Marks)		Bloom's Level	Course Outcome	Marks Allotted
1.	Write a regular expression for the language which accepts all strings with at least two a's over the set $\Sigma = \{a, b\}$.	[BL2]	[CO1]	[2]
2.	Construct a regular expression for the language over the set $\Sigma = \{0, 1\}$ in which total number of 1's are divisible by 3.	[BL2]	[CO1]	[2]
3.	Construct a regular expression denoting the set of all strings over $\{a, b\}$ such that all starts with a and ends with ab.	[BL2]	[CO1]	[2]
4.	Show that $(\phi)^* = \epsilon$ for a regular expression.	[BL1]	[CO1]	[2]
5.	Construct finite automata for the regular expression $(0+1)^*1^*$.	[BL2]	[CO1]	[2]
6.	Convert the following regular expressions to NFAs with ϵ - transitions $abb(a+b)^*$	[BL2]	[CO1]	[2]
7.	What are the algorithms of minimization DFA?	[BL1]	[CO1]	[2]
8.	State Pumping Lemma for regular Languages.	[BL1]	[CO1]	[2]
9.	Prove that the complement of a regular language is also regular.	[BL2]	[CO1]	[2]
10.	Find regular expression for the following DFA. 	[BL2]	[CO1]	[2]

PART B (13 Marks)

- i. Questions should be of Concept explanation, detailed reasoning, numerical problems, and derivations.
- ii. **Questions should be of Understanding / Applying/ Analyzing/ Evaluating**
- iii. Maximum number of Subdivisions is **Three**.

1.	i. Design a FA for the RE $(0+1)^*(00+11)01$. ii. Prove that the language $L = \{a^{m+n}b^ma^n \mid m, n \geq 1\}$ is not regular.	[BL4] [BL2]	[CO1] [CO1]	[7] [6]
2.	Compute Regular Expression for the given Finite Automata using formula method. 	[BL3]	[CO1]	[13]
3.	Find the regular expression for the following DFA using Arden's method. 	[BL3]	[CO1]	[13]

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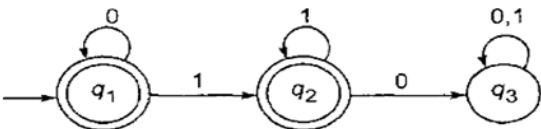
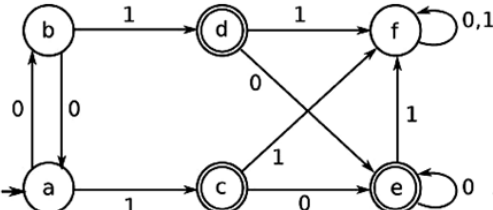
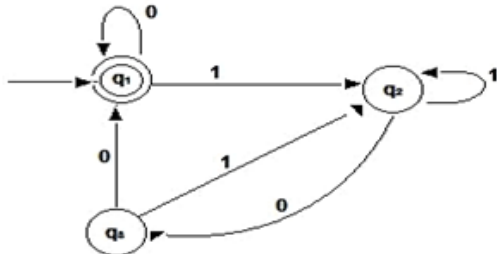
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4.	Construct a minimized DFA that can be derived from the following RE $a^*b(a ba)^*$.	[BL5]	[CO1]	[13]
5.	Construct a minimized DFA from the Regular expression $(x+y)y(x+y)^*$ and trace for a string $w=xyxyx$.	[BL5]	[CO1]	[13]
6.	Prove that the following language are not regular using pumping Lemma. i. $L=\{a^n b^n \mid n \geq 1\}$. ii. $L=\{a^p \mid p \text{ is a prime}\}$.	[BL2] [BL2]	[CO1] [CO1]	[7] [6]
7.	Show that the regular language is closed under: a. Union , b. Intersection c. Kleen closure d. Complement e. Concatenation f. Difference	[BL2]	[CO1]	[13]
8.	Explain in detail about closure properties of regular language.	[BL2]	[CO1]	[13]
9.	State and explain the conversion of DFA into RE using Arden's theorem. Illustrate with an example.	[BL5]	[CO1]	[13]
10.	Construct a NFA for regular expression $(b/a)^*baa$ and draw its equivalent minimized DFA.	[BL2]	[CO1]	[13]

Part C (15 Marks) (Only of Higher Order Thinking Question)

- Questions should be of Higher Order (HO) cognitive type.
- Formulate questions that align with the Evaluate and Create levels of higher-order thinking.**
- Maximum number of Subdivisions is **Three**.

1.	Convert the following DFA into Regular Expression using formula method. 	[BL5]	[CO1]	[15]
2.	Construct a minimized DFA. 	[BL6]	[CO1]	[15]
3.	Convert a finite automata to RE using Ardens method 	[BL6]	[CO1]	[15]
4.	Construct a minimized DFA from the Regular expression $(x+y)x(x+y)^*$. Trace for a string $w=xyxyx$.	[BL5]	[CO1]	[15]
5.	Is the set of strings over the alphabet $\{0\}$ of the form 0^n where n is not a prime is regular? Prove or disprove.	[BL6]	[CO1]	[15]

UNIT- III CONTEXT FREE GRAMMAR & PUSH DOWN AUTOMATA

Questions should be of:

- i. Basic concepts, definitions, laws, and simple applications.
- ii. Lower order (LO) cognitive and may be of Intermediate order (IO) cognitive type.

PART A (2 Marks)		Bloom's Level	Course Outcome	Marks Allotted
1.	Derive the string "00101" for rightmost derivation using a CFG given by, $S \rightarrow A1B$ $A \rightarrow 0A \mid \epsilon$ $B \rightarrow 0B \mid 1B \mid \epsilon$.	[BL2]	[CO2]	[2]
2.	Define Ambiguous grammar and CFG.	[BL1]	[CO2]	[2]
3.	Write the CFG for the language, $L=\{a^nba^n \mid n \geq 1\}$.	[BL2]	[CO2]	[2]
4.	Define pushdown automata.	[BL1]	[CO3]	[2]
5.	Express the ways of languages accepted by PDA and define them.	[BL2]	[CO3]	[2]
6.	State Pumping Lemma for CFL.	[BL1]	[CO2]	[2]
7.	Convert the following CFG to a push down automaton: $A \rightarrow 0A \mid 1A \mid 0 \mid 1$.	[BL2]	[CO3]	[2]
8.	Construct a PDA that accepts the language generated by the grammar. $S \rightarrow aSbb \mid aab$.	[BL2]	[CO3]	[2]
9.	Show that $L=\{a^p \mid P \text{ is prime}\}$ is not context free language.	[BL2]	[CO2]	[2]
10.	What are the closure properties of CFL? State the proof of any two properties.	[BL1]	[CO2]	[2]
PART B (13 Marks)				
i. Questions should be of Concept explanation, detailed reasoning, numerical problems, and derivations. ii. Questions should be of Understanding / Applying/ Analyzing/ Evaluating iii. Maximum number of Subdivisions is Three .				
1.	i. Solve the following grammar: $S \rightarrow aAa \mid bBb \mid BB$ $A \rightarrow C$ $B \rightarrow S \mid A$ $C \rightarrow S \mid \epsilon$ For the string abaabbba . Find left most, rightmost derivation and parse tree. ii. Show that the following grammar is ambiguous: $\{S \rightarrow a \mid S + S \mid S S \mid S^*S \mid (S)\}$.	[BL3]	[CO2]	[9]
2.	i. Construct a parse tree for the following grammar $G=(\{S, A\}, \{a, b\}, P, S)$ where P Consists of $S \rightarrow aAS \mid b, A \rightarrow SbA \mid ba$. Draw the derivation tree for the string $w=abbbab$. ii. Prove that the expression grammar is ambiguous $E \rightarrow E+E \mid E^*E \mid (E) \mid a$.	[BL3]	[CO2]	[9]
3.	Construct a PDA for language $L = \{0^{n+m}1^n2^m \mid n \geq 1, m \geq 1\}$.	[BL5]	[CO3]	[13]
4.	Construct the PDA for $L = \{WCW^R \mid W \text{ is in } (a+b)^+\}$.	[BL5]	[CO3]	[13]
5.	i. Construct Push down Automat to recognize the grammar G with following production and trace for a string of acceptance and rejection. $S \rightarrow aSA \mid \epsilon$ $A \rightarrow bB \mid cc$ $B \rightarrow bd \mid \epsilon$ ii. State pumping lemma for CFL. Use pumping lemma to show that the language $L = \{a^ib^jc^k \mid i < j < k\}$ is not a CFL.	[BL4]	[CO3]	[7]
6.	Construct the PDA for $L = \{WCW^R \mid W \text{ is in } (a+b)^+\}$.	[BL5]	[CO3]	[13]

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7.	Deduce PDA for the given CFG, and test whether 010^4 is acceptable by this PDA. $S \rightarrow 0BB$ $B \rightarrow 0S \mid 1S \mid 0.$	[BL4]	[CO3]	[13]
8.	Give a CFG for the language $N(M)$ where $M = (\{q_0, q_1\}, \{0, 1\}, \{z_0, x\}, \delta, q_0, z_0, \phi)$ where δ is given by, $\delta(q_0, 1, z_0) = \{(q_0, xz_0)\}$ $\delta(q_0, 1, x) = \{(q_0, xx)\}$ $\delta(q_0, 0, x) = \{(q_1, x)\}$ $\delta(q_0, \epsilon, z_0) = \{(q_0, \epsilon)\}$ $\delta(q_1, 1, x) = \{(q_1, \epsilon)\}$ $\delta(q_1, 0, z_0) = \{(q_0, z_0)\}.$	[BL5]	[CO3]	[13]
9.	i. Find out whether $L = \{x^n y n^2 n \mid n \geq 1\}$ is context free or not. ii. Let L be the language $\{ww \mid w \in \{0, 1\}^*\}$. Show that this language is not a CFL.	[BL3] [BL3]	[CO2] [CO2]	[6] [7]
10.	Describe in detail about closure properties of CFLs.	[BL2]	[CO2]	[13]
Part C (15 Marks) (Only of Higher Order Thinking Question) i. Questions should be of Higher Order (HO) cognitive type. ii. Formulate questions that align with the Evaluate and Create levels of higher-order thinking. iii. Maximum number of Subdivisions is Three .				
1.	Consider the following grammar $E \rightarrow E + T \mid T$ $E \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$ i. Give a rightmost derivation and leftmost derivation for the sentence $w = id * (id + id) * id$ ii. Is the Grammar Ambiguous? Justify iii. Construct the parse tree for the sentence in (i)	[BL5]	[CO2]	[15]
2.	Construct the PDA accepting the following languages i. $L = \{(ab)^n \mid n \geq 1\}$ by empty stack ii. $L = \{WWR \mid W \text{ is in } (a+b)^*\}$	[BL6]	[CO3]	[15]
3.	i. Show that the following grammars are Ambiguous $\{S \rightarrow aSbS \mid bSaS \mid \epsilon\}$ and $\{S \rightarrow AB \mid aaB, A \rightarrow a \mid Aa, B \rightarrow b\}$ ii. Find PDA that accept the given CFG $S \rightarrow XaaX$ $X \rightarrow ax \mid bx \mid \epsilon$	[BL5] [BL5]	[CO2] [CO3]	[7] [8]
4.	How PDA is converted into CFG? Convert the following PDA into CFG? $P = (\{p, q\}, \{0, 1\}, \{Z, X\}, \delta, p, Z, \phi)$ $\delta(q_0, 0, z_0) = \{(q_0, xz_0)\}$ $\delta(p, 1, Z) = \{(p, XZ)\}$ $\delta(p, \epsilon, Z) = \{(p, \epsilon)\}$ $\delta(p, 1, X) = \{(p, XX)\}$ $\delta(q, 1, x) = \{(q, \epsilon)\}$ $\delta(p, 0, X) = \{(q, X)\}$ $\delta(q, 0, Z) = \{(p, Z)\}$	[BL6]	[CO3]	[15]
5.	Construct a context free grammar for the languages $L(G_1) = \{a^i b^{2i} \mid i > 0\}$ and $L(G_2) = \{a^n b a^n \mid n > 0\}$	[BL6]	[CO2]	[15]

UNIT- IV - PROPERTIES OF CONTEXT-FREE LANGUAGES

Questions should be of:

- iii. Basic concepts, definitions, laws, and simple applications.
- iv. Lower order (LO) cognitive and may be of Intermediate order (IO) cognitive type.

PART A (2 Marks)		Bloom's Level	Course Outcome	Marks Allotted
1.	Convert the following grammar into an equivalent one with no unit productions and no useless symbols: $S \rightarrow ABA$ $A \rightarrow aAA aBC bB$ $B \rightarrow A bB Cb$ $C \rightarrow CC cC$	[BL2]	[CO3]	[2]
2.	Construct CFG without ϵ production from : $S \rightarrow a Ab aBa$, $A \rightarrow b \epsilon$, $B \rightarrow b A$.	[BL2]	[CO3]	[2]
3.	Convert the following grammar into an equivalent one with no unit productions, no useless symbols. $S \rightarrow A CB$ $A \rightarrow C D$ $B \rightarrow 1B 1$ $C \rightarrow 0C 0$ $D \rightarrow 2D 2$.	[BL2]	[CO3]	[2]
4.	State the two normal forms and give an example.	[BL1]	[CO3]	[2]
5.	Define GNF with an example.	[BL1]	[CO3]	[2]
6.	Define a Turing machine.	[BL1]	[CO4]	[2]
7.	Write the difference between finite automata and Turing machine.	[BL2]	[CO4]	[2]
8.	Design a TM that accepts the language of even integers written in binary.	[BL2]	[CO4]	[2]
9.	Construct a Turing machine for computing 2's complement of a binary number with the transition diagram.	[BL2]	[CO5]	[2]
10.	List the Programming Techniques for Turing Machine.	[BL1]	[CO4]	[2]
PART B (13 Marks)				
iv. Questions should be of Concept explanation, detailed reasoning, numerical problems, and derivations.				
v. Questions should be of Understanding / Applying/ Analyzing/ Evaluating				
vi. Maximum number of Subdivisions is Three .				
1.	Convert the following grammar to Chomsky Normal Form. $S \rightarrow A AB0 A1A$ $A \rightarrow A0 \epsilon$ $B \rightarrow B1 BC$ $C \rightarrow CB CA 1B$	[BL3]	[CO2]	[13]
2.	Convert the given CFG to CNF. Consider the given grammar G: $S \rightarrow a aA B$ $A \rightarrow aBB \epsilon$ $B \rightarrow Aa b$	[BL3]	[CO2]	[13]
3.	i. Obtain the Greibach Normal form equivalent to the grammar $S \rightarrow a AB$, $A \rightarrow a BC$, $B \rightarrow b$, $C \rightarrow b$. ii. Remove ϵ - production from the following grammar. $S \rightarrow ASA aB b$ $A \rightarrow B$, $B \rightarrow b \epsilon$.	[BL3] [BL3]	[CO2] [CO2]	[7] [6]

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4.	Express the following grammar G into Greibach Normal Form(GNF) $S \rightarrow XB \mid AA$ $A \rightarrow a \mid SA$ $B \rightarrow b, X \rightarrow a$	[BL5]	[CO2]	[13]
5.	Design a Turing Machine to recognize the language $L = \{0^n 1^n 2^n \mid n \geq 1\}$.	[BL4]	[CO4]	[13]
6.	Design a Turing Machine to perform proper subtraction.	[BL4]	[CO4]	[13]
7.	Design a Turing Machine for the language L that accepts all the strings with an equal number of a's and b's.	[BL4]	[CO4]	[13]
8.	Design a Turing Machine to recognize the following language $L = \{a^n b^n \mid n \geq 0\}$ and $L = \{1^n 0^n 1^n \mid n \geq 1\}$.	[BL4]	[CO4]	[13]
9.	Explain in detail about programming techniques of TM construction.	[BL2]	[CO5]	[13]
10.	Construct a Turing Machine for multiplying two non-negative integers using "copy" subroutine.	[BL4]	[CO5]	[13]

Part C (15 Marks) (Only of Higher Order Thinking Question)

i. Questions should be of Higher Order (HO) cognitive type.

ii. **Formulate questions that align with the Evaluate and Create levels of higher-order thinking.**

iii. Maximum number of Subdivisions is **Three**.

1.	What is the purpose of normalization? Construct the CNF and GNF for the following grammar and explain the steps. $S \rightarrow aAa \mid bBb \mid \epsilon$ $A \rightarrow C \mid a$ $B \rightarrow C \mid b$ $C \rightarrow CDE \mid \epsilon$ $D \rightarrow A \mid B \mid ab$	[BL5]	[CO2]	[15]
2.	State pumping lemma for CFL. Use pumping lemma to show that the language $L = \{a^i b^j c^k \mid i < j < k\}$ is not a CFL.	[BL6]	[CO3]	[15]
3.	Construct the TM which recognizes the language $L = \{WCW \mid W \in (a+b)^+\}$.	[BL5]	[CO4]	[15]
4.	Obtain a Greibach normal form grammar equivalent to the context free grammar $S \rightarrow 0 \mid AA$ $A \rightarrow 1 \mid SS$.	[BL5]	[CO2]	[15]
5.	Construct Turing machine for the language $\{WW \mid W \in \{a,b\}^*\}$.	[BL5]	[CO4]	[15]

UNIT- V UNDECIDABILITY

Questions should be of:

- i. Basic concepts, definitions, laws, and simple applications.
- ii. Lower order (LO) cognitive and may be of Intermediate order (IO) cognitive type.

PART A (2 Marks)		Bloom's Level	Course Outcome	Marks Allotted															
1.	When a language is said to be recursive or recursively language?	[BL2]	[CO6]	[2]															
2.	Let L be a language and L' be its complement. Identify whether the given language is recursive or recursively enumerable.	[BL2]	[CO6]	[2]															
3.	Let $\Sigma=\{0,1\}$. Let A and B be the lists of three strings each, defined as <table><tr><td></td><td>List A</td><td>List B</td></tr><tr><td>I</td><td>w_i</td><td>x_i</td></tr><tr><td>1</td><td>1</td><td>111</td></tr><tr><td>2</td><td>10111</td><td>10</td></tr><tr><td>3</td><td>10</td><td>0</td></tr></table> Does this PCP have a solution? Explain?		List A	List B	I	w_i	x_i	1	1	111	2	10111	10	3	10	0	[BL2]	[CO6]	[2]
	List A	List B																	
I	w_i	x_i																	
1	1	111																	
2	10111	10																	
3	10	0																	
4.	What is Post Correspondence Problem (PCP)?	[BL1]	[CO6]	[2]															
5.	Define Halting Problem.	[BL1]	[CO6]	[2]															
6.	Define Modified Post's correspondence problem.	[BL1]	[CO6]	[2]															
7.	Compare and contrast the language Ld.	[BL2]	[CO6]	[2]															
8.	When a language is said to recursively enumerable?	[BL2]	[CO6]	[2]															
9.	What is Universal Turing machine?	[BL1]	[CO6]	[2]															
10.	When we say a problem is decidable? Give an example of undecidable problem.	[BL2]	[CO6]	[2]															
PART B (13 Marks)																			
i. Questions should be of Concept explanation, detailed reasoning, numerical problems, and derivations.																			
ii. Questions should be of Understanding / Applying/ Analyzing/ Evaluating																			
iii. Maximum number of Subdivisions is Three.																			
1.	Find whether the following languages are recursive or recursively enumerable. i. Union of two recursive languages. ii. Intersection of two recursively enumerable languages. iii. Universal language Lu.	[BL4] [BL4] [BL4]	[CO6] [CO6] [CO6]	[4] [4] [5]															
2.	i Explain in detail about undecidable problems about Turing machines. ii. Prove that there exists a recursively enumerable language whose complement is not recursively enumerable.	[BL2] [BL4]	[CO6] [CO6]	[7] [6]															
3.	i. Define a language Ld and show that the language is not recursively enumerable. ii. State the halting problem of TMs. Prove that the halting problem of Turing Machine over $\{0,1\}^*$ as unsolvable.	[BL2] [BL2]	[CO6] [CO6]	[6] [7]															
4.	Prove that if L1 and L2 are Recursively Enumerable language over Σ , then $L1 \cup L2$ and $L1 \cap L2$ are also Recursively Enumerable.	[BL4]	[CO6]	[13]															
5.	Describe about Halting Problem in detail with suitable example.	[BL2]	[CO6]	[13]															
6.	i) Prove that if there is a reduction from P1 to P2, then if P1 is undecidable, then So is P2 (ii) Let $\Sigma = \{0, 1\}$ and A and B be the list as	[BL4]	[CO6]	[13]															

	<table><tr><td></td><td>List A</td><td>List B</td></tr><tr><td>i</td><td>w_i</td><td>x_i</td></tr><tr><td>1</td><td>10</td><td>101</td></tr><tr><td>2</td><td>011</td><td>11</td></tr><tr><td>3</td><td>101</td><td>011</td></tr></table>		List A	List B	i	w _i	x _i	1	10	101	2	011	11	3	101	011				
	List A	List B																		
i	w _i	x _i																		
1	10	101																		
2	011	11																		
3	101	011																		
Analyze the instance of PCP.																				
7.	i Explain in detail about undecidable problems about Turing machines. ii. Prove that there exists a recursively enumerable language whose complement is not recursively enumerable.	[BL2] [BL4]	[CO6] [CO6]	[7] [6]																
8.	State and Prove Post's correspondence problem.	[BL4]	[CO6]	[13]																
9.	Let $\Sigma=\{a,b\}^*$. Let A and B the lists of three strings as given below: A = {b, bab3, ba} B = {b3, ba, a} Does this instance of PCP have a solution? Justify your answer.	[BL5]	[CO6]	[13]																
10	Consider the turing machine M and w=01, where M = ({q1, q2, q3}, {0,1}, {0,1, B}, δ , q1, B, {q3}) and δ is given by <table><tr><td>q₁</td><td>$\delta(q,0)$</td><td>$\delta(q,1)$</td><td>$\delta(q,B)$</td></tr><tr><td>$\rightarrow q_1$</td><td>(q₂,1,R)</td><td>(q₂,0,L)</td><td>(q₂,1,L)</td></tr><tr><td>q₂</td><td>(q₃,0,L)</td><td>(q₁,0,R)</td><td>(q₂,0,R)</td></tr><tr><td>*q₃</td><td>-</td><td>-</td><td>-</td></tr></table> Reduce the above problem to Post correspondence problem and find whether that PCP has a solution or not.	q ₁	$\delta(q,0)$	$\delta(q,1)$	$\delta(q,B)$	$\rightarrow q_1$	(q ₂ ,1,R)	(q ₂ ,0,L)	(q ₂ ,1,L)	q ₂	(q ₃ ,0,L)	(q ₁ ,0,R)	(q ₂ ,0,R)	*q ₃	-	-	-	[BL5]	[CO6]	[13]
q ₁	$\delta(q,0)$	$\delta(q,1)$	$\delta(q,B)$																	
$\rightarrow q_1$	(q ₂ ,1,R)	(q ₂ ,0,L)	(q ₂ ,1,L)																	
q ₂	(q ₃ ,0,L)	(q ₁ ,0,R)	(q ₂ ,0,R)																	
*q ₃	-	-	-																	
Part C (15 Marks) (Only of Higher Order Thinking Question) i. Questions should be of Higher Order (HO) cognitive type. ii. Formulate questions that align with the Evaluate and Create levels of higher-order thinking. iii. Maximum number of Subdivisions is Three .																				
1.	Analyze the Halting Problem and prove that it is undecidable.	[BL5]	[CO6]	[15]																
2.	Evaluate the significance of Post Correspondence Problem in computability theory.	[BL5]	[CO6]	[15]																
3.	Analyze the relationship between PCP and Modified PCP.	[BL6]	[CO6]	[15]																
4.	Develop a Turing Machine for a recursively enumerable language and analyze its behavior.	[BL6]	[CO6]	[15]																
5.	Develop an algorithm to distinguish recursive and recursively enumerable languages.	[BL6]	[CO6]	[15]																